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Technique for GaN epitaxy on insulating substrates

Abstract

A method includes depositing a first epitaxial layer of an aluminum gallium nitride (AlGa_N) material onto a preliminary substrate and polishing the first layer's surface. Ions are implanted beneath the surface, which is bonded to a seed insulating substrate. Annealing is performed, resulting in second epitaxial layer on preliminary substrate and third epitaxial layer on seed insulating substrate. Third layer's surface is polished to obtain a seed wafer. In some implementations, a fourth epitaxial layer of a second AlGa_N material is deposited onto surface of third layer. Fourth layer's surface is polished, and ions are implanted beneath the surface, which is bonded to a product insulating substrate. Annealing is performed, resulting in fifth epitaxial layer on seed insulating substrate and sixth epitaxial layer on product insulating substrate. The sixth layer can be used to obtain an AlGa_N product, and the fifth layer can be reused to fabricate additional AlGa_N products.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Application No. 63/145,008, filed Feb. 3, 2021, which is hereby incorporated by reference.

BACKGROUND

(1) Gallium nitride (GaN) devices have many advantages over silicon devices, including lower on-resistance, faster switching, lower power, and smaller area. GaN devices fabricated on insulating substrates such as polycrystalline aluminum nitride (p-AlN) wafers can surprisingly have many advantages over GaN devices fabricated on Si even lower dislocation densities. The reason for these advantages is that AlGaN is first epitaxially grown on advantageous substrate such as single crystal sapphire, GaN, SiC or AlN. Using a multiple step process an epitaxial AlGaN layers are removed from initial wafers using an ion-implantation layer splitting process using one or more substrate layers, with additional CMP and epitaxial AlGaN growth steps and film transfer steps to place the epitaxial layers on the advantageous polycrystalline AlN substrate. One advantage is GaN devices on insulating substrates do not experience vertical breakdown voltages or backgating, and multiple GaN devices can be integrated into a single semiconductor die, reducing the area occupied by the GaN devices. Another advantage is that the final epitaxial layer thicknesses can be reduced creating a potential cost advantage of requiring less deposition per device wafers

(2) In addition, some techniques for GaN epitaxy on insulating substrates require the epitaxial layer to be transferred to several different carrier wafers in order to obtain a surface of the epitaxial layer with the desired face. For example in one technique, a gallium-facing GaN epitaxial layer is grown on a first wafer and then bonded to a temporary wafer following chemical mechanical polishing (CMP) process. The GaN epitaxial layer is then separated from the first wafer to expose a nitride-facing surface, which undergoes CMP. The nitride-facing surface is then bonded to the final wafer, and the GaN epitaxial layer is separated from the temporary wafer to expose the gallium-facing surface, which undergoes CMP. Any problems or advantages introduced in the GaN epitaxial layer during the two bonding and separation steps, as well as the CMP steps, are carried through to the final GaN device.

SUMMARY

(3) A method of epitaxial deposition includes depositing a first epitaxial layer of an aluminum gallium nitride (AlGa_N) material onto a preliminary substrate and polishing the first layer's surface. Ions are implanted at a depth beneath the surface of the first layer, which is bonded to a seed insulating substrate. Annealing is performed, such that the first epitaxial layer divides into a second epitaxial layer on the preliminary substrate and a third epitaxial layer on the seed insulating substrate. The third layer's surface is polished to obtain a seed wafer. The preliminary substrate can be chosen based on the AlGa_N material to reduce a dislocation density of the first epitaxial layer.

(4) The depth beneath the surface of the first epitaxial layer can be chosen based on an intended thickness of the third epitaxial layer on the seed insulating substrate. In some implementations, the second epitaxial layer is removed from the preliminary substrate such that the preliminary substrate can be reused. The preliminary substrate can be a sapphire wafer. The surface of the first epitaxial layer can be gallium facing, and the surface of the third epitaxial layer is nitride facing, in some implementations. The seed insulating substrate can be a polycrystalline aluminum nitride wafer.

(5) In some implementations, the AlGa_N material is a first AlGa_N material, and a fourth epitaxial layer of a second AlGa_N material is deposited onto the surface of the third epitaxial layer on the seed insulating substrate. The fourth layer's surface is polished, and ions are implanted at a depth beneath the surface, which is bonded to a product insulating substrate. Annealing is performed, such that the fourth epitaxial layer divides into a fifth epitaxial layer on the seed insulating substrate and a sixth epitaxial layer on the product insulating substrate. The surface of the sixth epitaxial layer is polished, and a seventh epitaxial layer of a third AlGa_N material is deposited onto the surface of the sixth epitaxial layer, resulting in an AlGa_N product.

(6) The product insulating substrate can be a same material as the seed insulating substrate. In some implementations, the depth beneath the surface of the fourth epitaxial layer is chosen based on an intended thickness of the sixth epitaxial layer. The surface of the third epitaxial layer is nitride facing, and the surface of the sixth epitaxial layer is gallium facing, in some implementations. The product insulating substrate is a polycrystalline aluminum nitride wafer in some implementations. The thickness of the fourth epitaxial layer can be chosen such that the fourth epitaxial layer can be used to fabricate a plurality of AlGa_N products.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. 1A-E illustrate an example fabrication process for a seed wafer.

(2) FIGS. 2A-E illustrate an example gallium nitride epitaxy process using the seed wafer shown in FIGS. 1A-E.

(3) FIGS. 3A-I illustrate a further example gallium nitride epitaxy process using the seed wafer shown in FIGS. 1A-E.

(4) FIG. 4 shows, in flowchart form, an example process for fabricating a seed wafer.

(5) FIG. 5 shows, in flowchart form, an example process for fabricating a product wafer.

DETAILED DESCRIPTION

(6) The same reference number is used in the drawings for the same or similar (either by function and/or structure) features.

(7) The described processes for aluminum gallium nitride (AlGaN) epitaxy include fabricating a preliminary wafer, depositing a first AlGaN epitaxial layer on the preliminary wafer, and fabricating a number of AlGaN seed wafers using the preliminary wafer and the first AlGaN epitaxial layer. The preliminary wafer includes a preliminary insulating wafer and a base epitaxial layer. The preliminary wafer is fabricated using substrates such as single crystal sapphire, GaN, silicon carbide (SiC), aluminum nitride (AlN), and the like. The thickness of the first AlGaN epitaxial layer is chosen based on an intended thickness of a resulting AlGaN epitaxial layer a AlGaN seed wafer, an amount removed by the chemical-mechanical polishing (CMP) steps and a number of AlGaN seed wafers to be fabricated. The number of AlGaN seed wafers are fabricated while the thickness of the first AlGaN epitaxial layer is greater than the intended thickness of the resulting AlGaN epitaxial layer.

(8) Fabricating an AlGaN seed wafer includes polishing a surface of the first AlGaN epitaxial layer and implanting ions at a depth beneath the surface of the first AlGaN epitaxial layer(s) based on the intended thickness of the resulting AlGaN epitaxial layer(s). The choice of ions, energy of ions, concentration of ions, wafer temperature of implantation are chosen based on the desired implementation. The surface of the first AlGaN epitaxial layer is bonded to a substrate that acts as the base of a seed wafer, such as polycrystalline AlN or SiC. After epitaxial deposition, CMP for planarization, bonding layer deposition, ion implantation and bonding to the seed insulating substrate, annealing is performed such that the first AlGaN epitaxial layer splits at the depth beneath the surface of the first AlGaN epitaxial layer. The resulting surfaces are smoothed and planarized using CMP.

(9) The seed wafers can then be used as a seed layer for a second AlGaN epitaxial layer having a composition and thickness chosen based on the desired characteristics of the resulting AlGaN product wafers. The second epitaxial layer is deposited on the seed layer followed by CMP, bonding layer deposition, ion implantation, bonding to a product insulating wafer, and separation with annealing. Both wafer surfaces undergo CMP, resulting in a seed wafer with a thinner first AlGaN epitaxial layer and a product wafer which can be used for additional epitaxial deposition and patterning.

(10) FIGS. 1A-E illustrate an example fabrication process **100** for a seed wafer. For ease of explanation, FIGS. 1A-E are described herein with reference to the example process **400** shown in FIG. 4 for fabricating a seed wafer. FIG. 1A illustrates a first step **100A** in the process **100** and step **410** in process **400**, in which an epitaxial layer **110** of a gallium nitride (AlGaN) material is deposited on a preliminary substrate **105**. The seed epitaxial layer **110** can be a multi-layer stack in some implementations. In some implementations, the epitaxial layer **110** is chosen to promote high-quality crystal growth and reduce the likelihood of defects such that the final product wafer performance and yield is increased. The preliminary substrate **105** is chosen to facilitate the growth of the epitaxial layer **110** such as a single crystal sapphire, GaN, SiC or AlN. The epitaxial AlGaN layer can be a complex multilayer chosen based on the preliminary substrate **105**. For example, sapphire can be the preliminary substrate **105** and an initial AlN film is deposited, as well as one of more layers of AlGaN. In some implementations, less AlN is used the farther from the preliminary substrate **105**.

(11) Although the product wafers include high quality GaN layers, the AlGaN layers on the initial seed layer of preliminary substrate **105** need not be the high quality GaN included in the GaN device layers. The thickness of the epitaxial layer **110** can be chosen based on the particular specification for the resulting seed wafer, and may range from 2 to 12 micrometers (μm). For example, a sapphire preliminary substrate **105** may be chosen to promote the growth of a two μm

gallium-facing (i.e., gallium polar) aluminum gallium nitride (AlGa_N) epitaxial layer **110**. The surface of the epitaxial layer **110** undergoes chemo-mechanical polishing (CMP) at step **420** of process **400** prior to bonding. The preliminary substrate **105** can be conditioned to promote seed layer growth. For example, a specific off-axis orientation can be chosen and/or controlled anneals performed to prepare the starting surface of preliminary substrate **105** for seed layer growth.

(12) In a second step **100B** shown in FIG. **1B** and step **430** of process **400**, ions **115** are implanted in the epitaxial layer **110** at a position chosen based on the desired thickness of the epitaxial layer **110** on the resulting seed wafer. The choice of ions, energy of ions, concentration of ions, wafer temperature of implantation and the like can be chosen based on the desired implementation. For example, hydrogen ions **115** may be implanted one half μm to one μm beneath a surface of the epitaxial layer **110**.

(13) FIG. **1C** illustrates a third step **100C**, in which a layer **120A** of a bonding agent is applied to the surface of the epitaxial layer **110** and/or a layer **120B** of the bonding agent is applied to the surface of a seed wafer **125**. The bonding agent can be silicon dioxide (SiO₂). The seed wafer **125** can be dense and provide a void-free surface. In some implementations, a material of the seed wafer **125** is chosen to have thermal expansion mismatch relative to the epitaxial layers. For example, p-AlN has a thermal expansion mismatch similar to AlN, AlGa_N, and GaN epitaxial layers. At step **440** of process **400**, the seed wafer **125** is bonded to the epitaxial layer **110** by the layers **120A** and/or **120B** of the bonding agent to obtain a single stage **130** including the preliminary substrate **105**, the epitaxial layer **110**, the implanted ions **115**, the bonding layer **120**, and the seed wafer **125**.

(14) In step **100D** illustrated in FIG. **1D** and step **450** of process **400**, the stage **130** is annealed such that the epitaxial layer **110** splits at the position of the implanted ions **115** to obtain two stages—a seed stage **135** and a preliminary stage **140**. The preliminary stage **140** includes the preliminary substrate **105** and a partial epitaxial layer **110A** from the epitaxial layer **110**, and at step **470** of process **400**, the preliminary stage **140** may be processed to remove the partial epitaxial layer **110A** such that the preliminary substrate **105** can be reused. Alternatively, the surface of the partial epitaxial layer **110A** can be smoothed using CMP and reused. The seed stage **135** includes the seed wafer **125**, the bonding layer **120**, and a partial epitaxial layer **110B** from the epitaxial layer **110**. In the example in which the hydrogen ions **115** are implanted one half μm beneath the surface of the epitaxial layer **110**, the partial epitaxial layer **110B** is one half μm thick. Both the preliminary stage **140** and the seed stage **135** can optionally be annealed to improve bonding or the quality of the epitaxial layers **110A** and **110B**.

(15) In some implementations in which the epitaxial layer **110** is AlGa_N, the partial epitaxial layer **110A** may be gallium-facing, and the partial epitaxial layer **110B** may be nitride-facing (i.e., nitride polar). The surface of the partial epitaxial layer **110B** of the stage **135** undergoes CMP in step **100E** illustrated in FIG. **1E** and step **460** of process **400**, resulting in the seed wafer **135** that can be used and reused in subsequent product fabrication to promote aligned deposition of other AlGa_N materials.

(16) FIGS. **2A-E** illustrate an example gallium nitride epitaxy process **200** using a seed wafer **135** from the process **100** shown in FIGS. **1A-E**. For ease of illustration, FIGS. **2A-E** are described herein with reference to the example process **500** shown in FIG. **5** for fabricating a product wafer. FIG. **2A** illustrates a first step **200A** in the process **200** and step **510** of process **500**, in which an epitaxial layer **210B** of an AlGa_N material is deposited on the seed wafer **235**. The seed wafer **235** is similar to the seed wafer **135** shown in FIG. **1E** and includes the seed wafer **225**, the bonding layer **220**, and the partial epitaxial layer **210A**. The AlGa_N material in epitaxial layer **210B** can be the same or different compared to the AlGa_N material in epitaxial layer **210A** of the seed wafer **235**.

(17) The thickness of the epitaxial layer **210B** can be chosen based on the particular specifications for the resulting product, and may range from 0.5 to 10 μm . For example, a one μm AlGa_N

epitaxial layer **210B** may be deposited on a nitride-facing AlGaIn epitaxial layer **210A**. The thickness of the epitaxial layer **210B** can be thinner than a thickness of an epitaxial layer **210A** deposited directly on the seed wafer **225**. The surface of the epitaxial layer **210B** undergoes CMP at step **520** of process **500**.

(18) In a second step **200B** shown in FIG. 2B and step **530** of process **500**, ions **215** are implanted in the epitaxial layer **210B** at a position chosen based on the desired thickness of the epitaxial layer **210** on the resulting product. For example, hydrogen ions may be implanted one half μm beneath a surface of the epitaxial layer **210B**. FIG. 2C illustrates a third step **200C**, in which a layer **240** of a bonding agent is applied to the surface of the epitaxial layer **210B** and/or the surface of a product wafer **245**. The material of the product wafer **245** can be the same or different from the material of the seed wafer **225** in the seed wafer **235**, and can be an insulating substrate with high thermal conductivity and a thermal expansion match to AlGaIn epitaxial layers. In some examples, the product wafer **245** is a p-AlN wafer.

(19) The bonding layer **240** can be the same or a different bonding agent as bonding layer **220** in seed wafer **235**. At step **540** of process **500**, the product wafer **245** is then bonded to the epitaxial layer **210B** by bonding layer **240** to obtain a single stage **250** including the seed wafer **225**, the bonding layer **220**, the epitaxial layers **210A** and **210B**, the implanted ions **215**, the bonding layer **240**, and the product wafer **245**. In step **200D** illustrated in FIG. 2D and step **550** of process **500**, the stage **250** is annealed such that the epitaxial layer **210B** splits at the position of the implanted ions **215** to obtain two stages—a product stage **255** and a seed wafer stage **260**.

(20) The seed wafer stage **260** includes the seed wafer **225**, the bonding layer **220**, and a partial epitaxial layer **210C**. The surface of the partial epitaxial layer **210C** undergoes CMP at step **590** of process **500**, and seed wafer stage **260** may be reused as a seed wafer **235** in subsequent performances of the process **200**. The product stage **255** includes the product wafer **245**, the bonding layer **240**, and the partial epitaxial layer **210D**. In the example in which the hydrogen ions **215** are implanted one half μm beneath the surface of the epitaxial layer **210B**, the partial epitaxial layer **210D** is one half μm thick.

(21) In some implementations in which the epitaxial layer **210** is AlGaIn, the partial epitaxial layer **210D** may be gallium-facing, and the partial epitaxial layer **210C** may be nitrogen-facing. The surface of the partial epitaxial layer **210D** of the product stage **255** undergoes CMP at step **560** of process **500**, such that the partial epitaxial layer **210D** is one half μm thick minus an amount removed during CMP for planarization in the previous example. In step **200E** illustrated in FIG. 2E and step **570** of process **500**, one or more additional epitaxial layers **270** are deposited on the surface of the partial epitaxial layer **210D** to form the resulting product. The material in the epitaxial layer **270** may be a doped or undoped GaN material or another appropriate material.

(22) The thickness and material of the epitaxial layer **270** can be chosen based on the particular specifications for the resulting product, such as a depletion-mode GaN device or an enhancement-mode GaN high electron mobility transistor. Many product devices include high electron mobility transistors, which use a combination of AlGaIn, AlN, and doped and undoped GaN for the epitaxial layer **270**. In addition, some product devices include multiple additional epitaxial layers **270** and a corresponding number of epitaxial deposition steps. For example, a one half μm unintentionally doped (UID) GaN epitaxial layer **270** may be deposited on the surface of the partial epitaxial layer **210D**.

(23) The process **200** described in FIGS. 2A-E reduces the number of bonding and debonding steps compared to some conventional processes for device fabrication and uses a thinner epitaxial layer **210** compared to some conventional processes, reducing the time required to deposit the epitaxial layer **210**. Although a thinner epitaxial layer reduces the time required to deposit the epitaxial layer, the time required to prepare the wafer for deposition and after the deposition remains the same.

(24) FIGS. 3A-J illustrate an example gallium nitride epitaxy process **300** for fabricating multiple devices using the seed wafer **135** shown in FIGS. 1A-E. The process **300** is similar to the process

200 but enables fabrication of multiple product wafers from a single epitaxial seed deposition, which reduces the total epitaxial processing time per product wafer. For ease of illustration, FIGS. **3A-J** are described herein with reference to the example process **500** shown in FIG. **5** for fabricating a product wafer. FIG. **3A** illustrates a first step **300A** in the process **300** and step **510** of process **500**, in which an epitaxial layer **310B** of an AlGa_N material is deposited on the seed wafer **335**. The seed wafer **335** is similar to the seed wafer **135** shown in FIG. **1E** and includes the seed wafer substrate **325**, the bonding layer **320**, and the partial epitaxial layer **310A**.

(25) The AlGa_N material in epitaxial layer **310B** can be the same as the AlGa_N material in epitaxial layer **310A** of the seed wafer **335**. The thickness of the epitaxial layer **310B** can be chosen based on the particular specification for the resulting products and a number of products to be created from the epitaxial layer **310B** and may range from 1 to 20 μm , for example. For example, the resulting products require approximately 0.4 μm of the epitaxial layer **310B**, and ten products are to be created from the epitaxial layer **310B**. A 5 μm AlGa_N epitaxial layer **310B** may be deposited on a nitride-facing AlGa_N epitaxial layer **310A**. The time spent in preparing and depositing the epitaxial layer **310B** is spread across the number of products to be fabricated from the process **300**. The surface of the epitaxial layer **310B** undergoes CMP at step **520** of process **500**.

(26) In a second step **300B** shown in FIG. **3B** and step **530** of process **500**, ions **315A** are implanted in the epitaxial layer **310B** at a position chosen based on the desired thickness of the epitaxial layer **310** on the resulting product. For example, hydrogen ions may be implanted 0.4 μm beneath a surface of the epitaxial layer **310B**. FIG. **3C** illustrates a third step **300C**, in which a layer **340A** of a bonding agent is applied to the surface of the epitaxial layer **310B** and/or the surface of a product wafer **345A**. The material of the product wafer **345A** can be the same or a different composition from the material of the seed wafer substrate **325** in the seed wafer **335**, and can be an insulating substrate with high thermal conductivity and a thermal expansion match to AlGa_N epitaxial layers. In some examples, the product wafer **345A** is a p-AlN wafer. The bonding layer **340A** can be the same or a different bonding agent as bonding layer **320** in seed wafer **335**. At step **540** of process **500**, the product wafer **345A** is then bonded to the epitaxial layer **310B** by bonding layer **340A** to obtain a single stage **350A** including the seed wafer **325**, the bonding layer **320**, the epitaxial layers **310A** and **310B**, the implanted ions **315A**, the bonding layer **340A**, and the product wafer **345A**.

(27) In step **300D** illustrated in FIG. **3D** and step **550** of process **500**, the stage **350A** is annealed such that the epitaxial layer **310B** splits at the position of the implanted ions **315A** to obtain two stages—a product stage **355A** and a seed wafer stage **360A**. The seed wafer stage **360A** includes the seed wafer **325**, the bonding layer **320**, the epitaxial layer **310A**, and a partial epitaxial layer **310C**. The product stage **355A** includes the product wafer **345A**, the bonding layer **340A**, and the partial epitaxial layer **310D**. In some implementations in which the epitaxial layer **310** is AlGa_N, the partial epitaxial layer **310D** on the product stage **355A** may be gallium-facing, and the partial epitaxial layer **310C** on the seed wafer stage **360A** may be nitrogen-facing.

(28) The surface of the partial epitaxial layer **310C** on the seed wafer stage **360A** undergoes CMP at step **590** of process **500**. While the remaining epitaxial layer **310C** is greater than a threshold thickness as determined at step **580**, the remaining epitaxial layer **310C** and seed wafer stage **360A** are reused as a seed wafer **335** in subsequent performances of the steps **300B** and **300C** as discussed further herein with respect to FIGS. **3F** and **3G** to fabricate additional product stages **355**. In the example in which the epitaxial layer **310B** is 5 μm thick and the hydrogen ions **315A** are implanted 0.4 μm beneath the surface of the epitaxial layer **310B**, the partial epitaxial layer **310D** on the product stage **355A** is 0.4 μm thick, and the partial epitaxial layer **310C** on the seed wafer stage **360A** is approximately 4.6 μm thick. Approximately 0.1 μm of the partial epitaxial layer **310C** is removed during CMP, and the remaining 4.5 μm of the partial epitaxial layer **310C** on the seed wafer stage **360A** can be used to fabricate the remaining nine products.

(29) The surface of the partial epitaxial layer **310D** of the product stage **355A** undergoes CMP at step **560** of process **500**, and in step **300E** illustrated in FIG. **3E** and step **570** of process **500**, one or

more additional epitaxial layers **370A** are deposited on the surface of the partial epitaxial layer **310D** to form the resulting product. Many product devices include high electron mobility transistors, which use a combination of AlGa_N, AlN, and doped and undoped GaN for the epitaxial layer **370A**. In addition, some product devices include multiple additional epitaxial layers **370A** and a corresponding number of epitaxial deposition steps. The thickness of the epitaxial layer **370A** can be chosen based on the particular specifications for the resulting product. For example, a one half μm unintentionally doped (UID) GaN epitaxial layer **370A** may be deposited on the surface of the partial epitaxial layer **310D**.

(30) Step **300F** shown in FIG. **3F** is similar to step **300B** shown in FIG. **3B** and step **530** of process **500**, and the seed wafer stage **360A** is reused in the fabrication of a second product. Ions **315B** are implanted in the epitaxial layer **310C** at a position chosen based on the desired thickness of the epitaxial layer **310** on the resulting product. For example, hydrogen ions may be implanted $0.4\ \mu\text{m}$ beneath a surface of the epitaxial layer **310C**. The ions **315B** can be the same as or different than the ions **315A**, and implanted at a same or different depth beneath the surface of the epitaxial layer **310** as ions **315A**.

(31) Step **300G** shown in FIG. **3G** is similar to step **300C** shown in FIG. **3C**, and a layer **340B** of a bonding agent is applied to the surface of the epitaxial layer **310C** and/or the surface of a product wafer **345B**. The material of the product wafer **345B** can be the same as the material of the product wafer **345A** or the seed wafer substrate **325** in the seed wafer **335**. In some examples, the product wafer **345B** is a p-AlN wafer. The bonding layer **340B** can be the same or a different bonding agent as bonding layer **340A** or bonding layer **320** in seed wafer **335**. At step **540** of process **500**, the product wafer **345B** is then bonded to the epitaxial layer **310C** by bonding layer **340B** to obtain a single stage **350B** including the seed wafer **325**, the bonding layer **320**, the epitaxial layers **310A** and **310C**, the implanted ions **315B**, the bonding layer **340B**, and the product wafer **345B**.

(32) Step **300H** shown in FIG. **3H** is similar to step **300D** shown in FIG. **3D** and step **550** of process **500**, and the stage **350B** is annealed such that the epitaxial layer **310C** splits at the position of the implanted ions **315B** to obtain two stages—a product stage **355B** and a seed wafer stage **360B**. The seed wafer stage **360B** includes the seed wafer substrate **325**, the bonding layer **320**, the epitaxial layer **310A**, and a partial epitaxial layer **310E**. The product stage **355B** includes the product wafer **345B**, the bonding layer **340B**, and the partial epitaxial layer **310F**. In some implementations in which the epitaxial layer **310** is AlGa_N, the partial epitaxial layer **310F** on product stage **355B** may be gallium-facing, and the partial epitaxial layer **310E** on seed wafer stage **360B** may be nitrogen-facing.

(33) The surface of the partial epitaxial layer **310E** on seed wafer stage **360B** undergoes CMP at step **590** of process **500**. While the remaining epitaxial layer **310E** is greater than a threshold thickness as determined at step **580**, the remaining epitaxial layer **310E** and seed wafer stage **360B** are reused to fabricate other products. In the example in which the epitaxial layer **310C** is $4.6\ \mu\text{m}$ thick and the hydrogen ions **315B** are implanted $0.4\ \mu\text{m}$ beneath the surface of the epitaxial layer **310C**, the partial epitaxial layer **310F** on product stage **355B** is $0.4\ \mu\text{m}$ thick, and the partial epitaxial layer **310E** on the seed wafer stage **360B** is approximately $4.1\ \mu\text{m}$ thick. Approximately $0.1\ \mu\text{m}$ of the partial epitaxial layer **310E** is removed during CMP at step **590** of process **500**, and the remaining $4\ \mu\text{m}$ of the partial epitaxial layer **310E** on the seed wafer stage **360B** can be used to fabricate the remaining eight products.

(34) The surface of the partial epitaxial layer **310F** of the product stage **355B** undergoes CMP at step **560** of process **500**, and step **300I** illustrated in FIG. **3I** is similar to step **300E** illustrated in FIG. **3E** and step **570** of process **500**. One or more additional epitaxial layers **370B** are deposited on the surface of the partial epitaxial layer **310F** to form the resulting product. The material in the epitaxial layer **370B** may be a doped or undoped GaN material or another appropriate material. The thickness of the epitaxial layer **370B** can be chosen based on the particular specifications for the resulting product. For example, a one half μm unintentionally doped (UID) GaN epitaxial layer

370B may be deposited on the surface of the partial epitaxial layer **310F**.

(35) The process **300** described in FIGS. **3A-I** reduces the number of bonding and debonding steps compared to some conventional processes for product fabrication and reuses a single seed wafer with an epitaxial deposition for several products, to spread the time spent in preparing and depositing the epitaxial layer across the several products.

(36) Unless otherwise stated, “about,” “approximately,” or “substantially” preceding a value means ± 10 percent of the stated value. Modifications are possible in the described examples, and other examples are possible within the scope of the claims.

(37) Modifications are possible in the described embodiments, and other embodiments are possible, within the scope of the claims.

Claims

1. A method, comprising: forming a first epitaxial layer of a gallium nitride (GaN) material onto a first substrate; bonding a surface of the first epitaxial layer to a second substrate, the second substrate including an insulating material; splitting the first epitaxial layer into a second epitaxial layer on the first substrate and a third epitaxial layer on the second substrate; forming a first wafer including the third epitaxial layer on the second substrate; and forming a second wafer having a third substrate using the first wafer.
2. The method of claim 1, wherein the gallium nitride material is an aluminum gallium nitride (AlGaN) material, and the first substrate reduces a dislocation density of the first epitaxial layer.
3. The method of claim 1, further comprising providing ions at a depth beneath a surface of the first epitaxial layer, wherein the depth beneath the surface of the first epitaxial layer is based on a thickness of the third epitaxial layer.
4. The method of claim 1, further comprising removing the second epitaxial layer from the first substrate.
5. The method of claim 1, wherein the first substrate includes sapphire.
6. The method of claim 1, wherein the surface of the first epitaxial layer is gallium facing and a surface of the third epitaxial layer is nitride facing.
7. The method of claim 1, wherein the second substrate includes aluminum nitride.
8. The method of claim 1, wherein the GaN material is a first GaN material, and forming a second wafer having a third substrate using the first wafer includes: forming a fourth epitaxial layer of a second GaN material onto a surface of the third epitaxial layer of the first wafer; providing ions at a depth beneath a surface of the fourth epitaxial layer; bonding the surface of the fourth epitaxial layer to the third substrate, the third substrate including the insulating material; splitting the fourth epitaxial layer into a fifth epitaxial layer on the second substrate and a sixth epitaxial layer on the third substrate; and forming a seventh epitaxial layer of a third GaN material onto a surface of the sixth epitaxial layer on the third substrate to form the second wafer.
9. The method of claim 8, wherein the third substrate includes a same material as the second substrate.
10. The method of claim 8, wherein the depth beneath the surface of the fourth epitaxial layer is based on a thickness of the sixth epitaxial layer.
11. The method of claim 8, wherein the surface of the third epitaxial layer is nitride facing and the surface of the sixth epitaxial layer is gallium facing.
12. The method of claim 8, wherein the third substrate includes aluminum nitride.
13. The method of claim 8, further comprising: forming a plurality of GaN devices using the fourth epitaxial layer.
14. The method of claim 13, further comprising: providing ions at a depth beneath a surface of the fifth epitaxial layer; bonding the surface of the fifth epitaxial layer to a fourth substrate, the fourth substrate including the insulating material; splitting the fifth epitaxial layer into an eighth epitaxial

layer on the second substrate and a ninth epitaxial layer on the fourth substrate; and forming a tenth epitaxial layer of a fourth GaN material onto a surface of the ninth epitaxial layer on the fourth substrate to form a third wafer.

15. A method, comprising: forming a first epitaxial layer onto a base epitaxial layer of a first substrate; providing ions at a depth beneath a surface of the first epitaxial layer; bonding the surface of the first epitaxial layer to a second substrate, the second substrate including an insulating material; splitting the first epitaxial layer into a second epitaxial layer on the first substrate and a third epitaxial layer on the second substrate; forming a fourth epitaxial layer of a second material onto a surface of the third epitaxial layer on the second substrate to obtain a wafer; and forming devices in the fourth epitaxial layer of the wafer.

16. The method of claim 15, wherein the depth beneath the surface of the first epitaxial layer is based on a thickness of the third epitaxial layer.

17. The method of claim 15, wherein the first and second substrates include aluminum nitride.

18. The method of claim 15, further comprising forming a plurality of devices using the first epitaxial layer.

19. The method of claim 18, further comprising: providing ions at a depth beneath a surface of the second epitaxial layer; bonding the surface of the second epitaxial layer to a third substrate, the third substrate including the insulating material; splitting the second epitaxial layer into a fifth epitaxial layer on the first substrate and a sixth epitaxial layer on the third substrate; and forming a seventh epitaxial layer onto a surface of the sixth epitaxial layer on the third substrate.

20. A method, comprising: forming a first substrate and a base epitaxial layer on the first substrate; forming a first epitaxial layer over the base epitaxial layer; and providing ions at a depth beneath a surface of the first epitaxial layer; bonding the surface of the first epitaxial layer to a second substrate, the second substrate including an insulating material; splitting the first epitaxial layer at the depth beneath the surface of the first epitaxial layer to form a second epitaxial layer bonded to the second substrate; and forming a third epitaxial layer on a surface of the second epitaxial layer to form a wafer including the second substrate.

21. The method of claim 20, wherein forming a first substrate and a base epitaxial layer on the first substrate includes: forming a third epitaxial layer on a third substrate; providing ions at a depth beneath a surface of the third epitaxial layer; bonding the surface of the third epitaxial layer to the first substrate; and splitting the third epitaxial layer to form the base epitaxial layer bonded to the first substrate.

22. The method of claim 20, wherein the base epitaxial layer, the first epitaxial layer, and the second epitaxial layer each includes a respective GaN material.

23. The method of claim 20, wherein the first substrate includes the insulating material.

24. The method of claim 23, wherein the insulating material includes aluminum nitride.

25. The method of claim 15, wherein the first epitaxial layer and the base epitaxial layer are of an AlGaN material.

26. The method of claim 15, wherein providing ions at a depth beneath a surface of the first epitaxial layer includes implanting the ions at the depth.

27. The method of claim 15, wherein splitting the first epitaxial layer into a second epitaxial layer on the first substrate and a third epitaxial layer on the second substrate includes annealing the first epitaxial layer.

28. The method of claim 1, wherein providing ions at a depth beneath a surface of the first epitaxial layer includes implanting the ions at the depth.

29. The method of claim 1, wherein splitting the first epitaxial layer into a second epitaxial layer on the first substrate and a third epitaxial layer on the second substrate includes annealing the first epitaxial layer.

30. The method of claim 1, further comprising polishing the surface of the first epitaxial layer and polishing a surface of the third epitaxial layer.

31. The method of claim 1, further comprising forming devices on the second wafer.
 32. The method of claim 15, further comprising forming devices on the wafer.
 33. The method of claim 20, further comprising forming devices on the wafer.
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